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A Dissertation Proposal on
“Comparative Benchmarking of CatBoost, TabPFN, and TabFM for
Explainable Multi-Tier Hair Fall Risk Stratification”

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Submitted to
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In partial fulfillment of the requirement for Master’s Degree in Computer
Science and Information technology (M.Sc. CSIT), 4th semester

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List of Abbreviation

Abbreviation	Full Form
AI	Artificial Intelligence
ALT	Alanine Aminotransferase
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
CV	Cross-Validation
EHR	Electronic Health Records
GBDT	Gradient Boosted Decision Tree
GPU	Graphics Processing Unit
LTS	Long Term Support
ML	Machine Learning
MPS	Metal Performance Shaders
NeurIPS	Neural Information Processing Systems
NumPy	Numerical Python
PSS	Perceived Stress Scale
SHAP	Shapley Additive Explanations
RAM	Random Access Memory
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
XAI	Explainable Artificial Intelligence

TabPFN Tabular Prior-Fitted Network

TabFM TabFM

Chapter 1: Introduction

1.1 Overview

Alopecia, the clinical term for pathological hair loss, affects individuals across a broad range of ages and demographic backgrounds worldwide. Some degree of hair shedding is a normal part of the follicular growth cycle, but a marked or sudden increase in shedding often points to an underlying disturbance in the body's systems commonly hormonal imbalance, significant nutrient depletion, prolonged physical strain, or sustained psychological stress. Beyond its visible effects, noticeable hair loss frequently carries a real emotional cost, contributing to lowered self-confidence, elevated anxiety, and a reduced sense of overall well-being.

Determining an individual's likelihood of experiencing hair loss is inherently difficult because no single biomarker reliably predicts it. Risk instead emerges from the combined, non-linear interaction of physiological traits, everyday habits, environmental factors, and pre-existing health conditions. Elements such as diet quality, sleep regularity, scalp care practices, thyroid activity, iron levels, and hereditary predisposition all interact at once, making it impractical for clinicians or individuals to gauge overall risk through observation alone.

Healthcare systems today generate large volumes of patient data through Electronic Health Records (EHRs), capturing detailed clinical histories over time. Machine learning offers a way to surface patterns within this dense, multi-source data that would otherwise go unnoticed. Gradient boosting methods have long been the standard tool for prediction on structured, tabular health data. More recently, however, tabular foundation models transformer architectures pretrained to perform prediction directly from context, without task-specific training have begun to reshape this space. Google's TabFM and Prior Labs' TabPFN both represent this newer generation of zero-shot tabular models, offering predictions through a single forward pass rather than the conventional train-and-tune cycle.

A persistent limitation of both gradient boosting and transformer-based approaches, however, is their tendency to operate as opaque, uninterpretable systems. This research addresses that limitation by building a comparative machine learning framework for evaluating and explaining multi-tier hair fall risk, using structured clinical and lifestyle data. The study

benchmarks three distinct modeling paradigms: CatBoost, a gradient-boosted decision tree architecture purpose-built for categorical data; TabPFN, an established prior-fitted transformer foundation model for tabular in-context learning; and TabFM, Google Research's more recently released zero-shot tabular foundation model, which combines row-and-column attention with an in-context learning transformer. Post-hoc explainability is layered across all three models via Shapley Additive Explanations (SHAP), giving each prediction a transparent, feature-level justification grounded in the underlying medical and lifestyle indicators.

1.2 Problem Statement

Determining hair loss risk remains a genuinely difficult clinical problem, shaped by the interplay of hormonal shifts, metabolic irregularities, chronic stress, nutritional gaps, and hair care habits. Current practice leans heavily on subjective self-reporting or delayed clinical evaluation, both of which are vulnerable to observer bias, inconsistent standards, and limited reach in under-resourced settings.

While computational methods can, in principle, model these interacting risk factors systematically, most existing dermatological prediction research still relies on older algorithms Logistic Regression or standard Random Forests that demand substantial manual feature engineering and struggle to capture complex feature interactions. Moreover, the dermatology literature has not yet meaningfully compared how newer zero-shot tabular foundation models perform against specialized gradient-boosted trees on real-world health survey data, and no existing work has extended this comparison to include more than one such foundation model architecture.

This study addresses these gaps directly by building and benchmarking an explainable machine learning framework for multi-tier hair fall risk classification. By evaluating CatBoost against two independently developed tabular foundation models TabPFN and TabFM alongside SHAP-based feature attribution, this research examines whether zero-shot, prior-fitted transformer approaches can match or exceed a tuned gradient-boosted tree architecture on tabular health data, while keeping every predicted risk tier traceable to specific, interpretable clinical indicators.

1.3 Objective

The objective of this dissertation is:

- To design, benchmark, and explain a machine learning framework comparing CatBoost, TabPFN, and TabFM for predicting multi-tier hair fall risk from structured clinical and lifestyle data, while providing transparent, feature-level decision-support insights through SHAP-based explainability.

Chapter 2: Background Study and Literature Review

2.1 Background Study

Alopecia progresses through a range of clinical stages, from mild, barely noticeable shedding to significant, visible scalp thinning. Conventional clinical assessment depends largely on subjective observation and periodic self-reporting from patients, an approach that tends to be inconsistent, susceptible to observer bias, and difficult to access in under-resourced or rural areas.

Machine learning offers a more systematic alternative, capable of modeling the simultaneous influence of physiological, behavioral, and clinical variables on hair loss outcomes. This study draws on a dataset of 21,606 records combining demographic attributes, biomarker measurements, and lifestyle indicators, used to classify individuals into three ordinal risk tiers: Low, Moderate, and High. The dataset was constructed using domain-informed relationships between known dermatological risk factors such as iron deficiency, insufficient protein intake, and elevated stress and hair fall outcomes, rather than being sourced from a single pre-existing clinical registry.

2.1.1 Comparative AI & Explainability Architecture

This research builds an integrated pipeline that carries structured clinical data through four distinct architectural components:

1. **State-of-the-Art Categorical Boosting (CatBoost):** CatBoost is a gradient-boosted decision tree framework built specifically to handle tabular data with a high proportion of categorical variables. Rather than relying on One-Hot Encoding, which tends to introduce sparsity and target leakage in high-cardinality features, CatBoost applies Ordered Target Encoding, converting each categorical value into a continuous statistic derived from the target variable under a randomly permuted data ordering.

For a dataset permuted randomly by $\sigma = (\sigma_1, \dots, \sigma_n)$, the categorical feature value x_k for the k -th instance is transformed into a continuous target statistic $\hat{x}_k$ using prior smoothing:

$$\hat{x}_k = \frac{\sum_{j=1}^{k-1} [x_{\sigma_j} = x_{\sigma_k}] \cdot y_{\sigma_j} + a \cdot p}{\sum_{j=1}^{k-1} [x_{\sigma_j} = x_{\sigma_k}] + a}$$

where y_{σ_j} denotes the target risk level, p is a global prior value, and $a > 0$ is a smoothing weight parameter.

CatBoost additionally builds Oblivious (Symmetric) Decision Trees, applying an identical split condition across every node at a given tree depth d . The objective function minimizes regularized loss at iteration t :

$$\mathcal{L}^{(t)} = \sum_{i=1}^N L(y_i, F_{t-1}(x_i) + h_t(x_i)) + \Omega(h_t)$$

where, $L(\cdot)$ is the multi-class cross-entropy loss function and $\Omega(h_t)$ penalizes tree complexity. This structural symmetry acts as a strong regularizer against target shift, giving the model a built-in resistance to overfitting on relatively small clinical survey data.

2. **Frontier Tabular Foundation Model TabPFN:** TabPFN (Prior-Fitted Network) was among the first transformer architectures designed specifically for tabular prediction via in-context learning. Pretrained entirely offline on synthetic datasets generated from structural causal models, TabPFN produces predictions in a single forward pass, without any gradient-based training on the target dataset.

Given a training set $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n$ and an unlabeled query instance x_{test} , TabPFN passes the concatenated dataset directly into its self-attention mechanism, evaluating the posterior predictive distribution in a single forward pass without iterative backpropagation:

$$P(y_{\text{test}} | x_{\text{test}}, \mathcal{D}_{\text{train}}) = \int P(y_{\text{test}} | x_{\text{test}}, \theta) P(\theta | \mathcal{D}_{\text{train}}) d\theta$$

The attention mechanism computes contextual relationships across both feature dimensions and patient samples simultaneously using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

TabPFN scales this mechanism to accommodate larger sample sizes, allowing zero-shot inference over complex health indicators without manual hyperparameter tuning or feature scaling.

3. **Zero-Shot Tabular Foundation Model TabFM:** TabFM, introduced by Google Research in mid-2026, represents a more recent entry in the tabular foundation model space, combining architectural ideas from both TabPFN and TabICL. Where TabPFN applies attention primarily across the sample dimension, TabFM alternates row-wise and column-wise attention over the raw input table before compressing each row into a dense embedding, which a dedicated in-context transformer then reads to produce predictions.

Like TabPFN, TabFM requires no dataset-specific training or hyperparameter search: the fit step stores the training context rather than updating model weights, and inference proceeds through a single forward pass:

$$P(y_{\text{test}} \mid x_{\text{test}}, D_{\text{train}}) = \text{TabFM}(x_{\text{test}}, D_{\text{train}})$$

TabFM was pretrained on a large corpus of synthetically generated tabular datasets built from structural causal models, a design choice motivated by the limited availability of large-scale, privacy-cleared real-world tabular data suitable for pretraining at this scale. The model supports classification tasks with up to ten output classes and approximately 500 input features, comfortably accommodating the three-tier, twenty-feature structure used in this study.

4. **Explainability Framework (SHAP):** To address the black-box nature shared by boosting and transformer architectures alike, this study applies post-hoc Shapley Additive Explanations across all three models. Rooted in cooperative game theory, SHAP decomposes each model's output into additive per-feature

contributions (ϕ_i), quantifying how far each clinical or lifestyle variable shifts a prediction away from its baseline expected value (ϕ_0):

$$f(x) = \phi_0 + \sum_{j=1}^M \phi_j$$

The local marginal contribution ϕ_j for feature j across feature subsets $S \subseteq F \setminus \{j\}$ is computed mathematically as:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)]$$

This formulation yields both patient-level waterfall plots showing individual prediction drivers, and cohort-level rankings showing which features matter most across the dataset as a whole, applied consistently across CatBoost, TabPFN, and TabFM.

2.2 Literature Review

Machine learning approaches to hair loss prediction have grown steadily in recent years. Widowati et al. [1] developed an explainable decision-support system for multi-class hair loss severity classification, emphasizing that restricting feature transformations to within cross-validation folds prevents inflated generalization estimates.

Sayyad and Midhunchakkaravarthy [2] built a diagnostic model specifically targeting Alopecia Areata, improving prediction accuracy for that single condition, though their approach did not extend to the broader set of multi-factorial risk drivers relevant to hair loss generally.

Earlier comparative research [3] benchmarked individual classifiers against basic ensemble methods for hair fall prediction, finding that combining multiple decision boundaries produced more stable predictions than any single estimator alone.

A related study [4] proposed a Random Forest-based framework incorporating hereditary factors, hormonal markers, medical history, and nutritional status, achieving reasonable baseline accuracy but lacking any mechanism for explaining which variables drove individual predictions.

A separate deep learning investigation [5] combined genetic, hormonal, and scalp-condition data with lifestyle variables to compare standard classifiers against neural network architectures, noting that while neural models captured more complex feature interactions, this came at the cost of per-prediction interpretability.

Another applied study [6] examined survey-based lifestyle and clinical indicators, including dandruff severity, chronic illness history, thyroid function, and micronutrient status, using ROC-AUC and macro-F1 to account for class imbalance and emphasizing the importance of correct categorical feature handling for ordinal medical outcomes.

Recent dermatological and scalp-health classification studies [7] have shown CatBoost outperforming conventional level-wise tree algorithms on datasets dominated by categorical health survey features, a performance advantage attributed to CatBoost's Ordered Target Encoding and its resistance to target leakage.

The emergence of tabular foundation models such as TabPFN [8] has established a new benchmark for zero-shot clinical decision support, achieving competitive predictive performance without iterative hyperparameter tuning and making it particularly well suited to small- and medium-sized clinical datasets.

Finally, the broader clinical machine learning literature [9] stresses that predictive accuracy alone is insufficient without accompanying explainability, and that integrating SHAP-based

feature attributions with modern tabular models allows practitioners to cross-check automated predictions against known biological mechanisms.

Google Research's TabFM [10] extends this paradigm through a hybrid attention architecture that alternates row- and column-wise processing before applying in-context prediction, and has been shown on the TabArena benchmark suite to outperform heavily tuned gradient-boosted baselines in a purely zero-shot setting.

Chapter 3: Methodology

3.1 Research Framework

The methodology adopted in this research follows an end-to-end, multi-tier machine learning pipeline designed to train, evaluate, and interpret hair fall risk predictions. The overall workflow unfolds across four sequential stages: data ingestion and preprocessing, parallel model training and inference across three distinct architectures (CatBoost, TabPFN, and TabFM), quantitative performance evaluation, and post-hoc explainability through SHAP.

Raw patient records first pass through a preprocessing stage that removes non-informative identifier fields, encodes categorical attributes, and prepares the feature set

for downstream modeling. The cleaned data then branches into three independent modeling paths, each corresponding to one of the benchmarked architectures: CatBoost, trained through iterative gradient-boosted decision trees; TabPFN, operating through zero-shot in-context inference; and TabFM, Google's more recently introduced tabular foundation model, which likewise performs zero-shot prediction but through an alternating row-and-column attention mechanism. Each model's predictions are then passed through a shared evaluation stage, where accuracy, precision, recall, F1-score, and ROC-AUC are computed under a consistent methodology to allow direct comparison across all three approaches. Finally, SHAP is applied uniformly across all three trained models, producing both global, cohort-level feature importance rankings and individual, patient-level explanations for specific predictions.

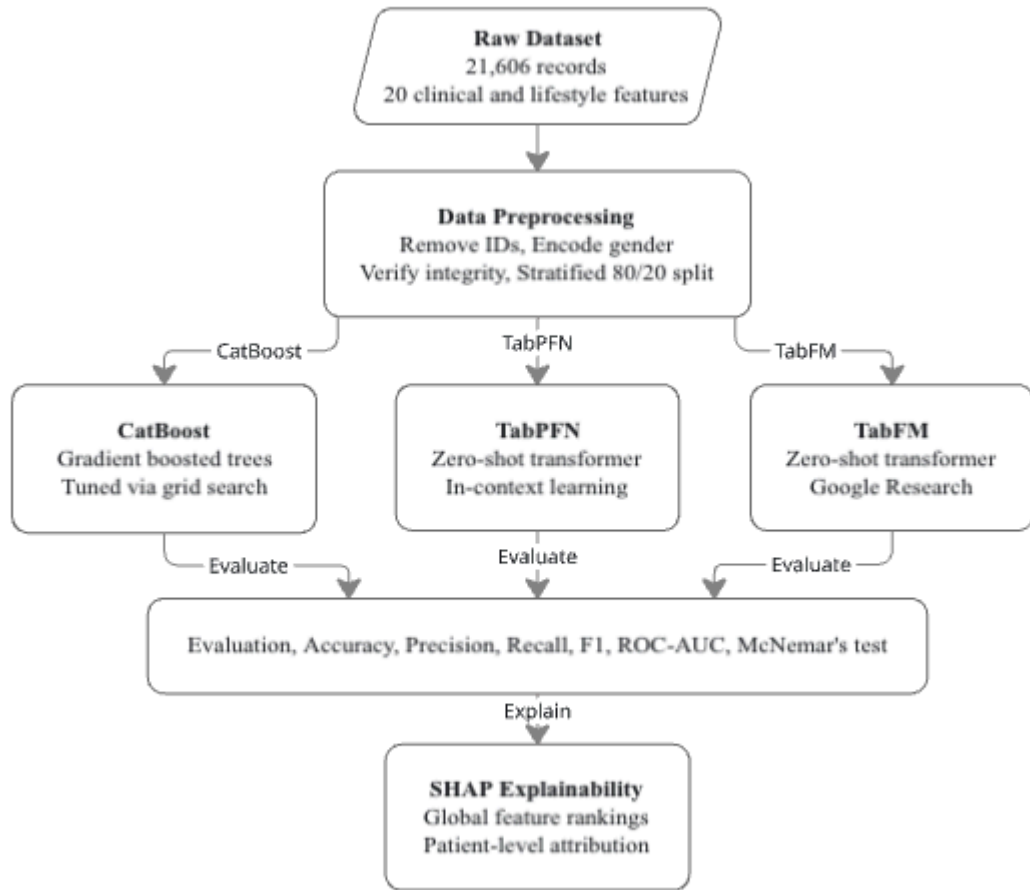


Figure 1: Overall System Architecture Diagram

3.2 Dataset Description

The dataset used in this study consists of 21,606 patient records, combining demographic attributes, continuous physiological biomarkers, and binary lifestyle and clinical indicators. The target variable, hair fall risk, is represented as three ordinal severity tiers: Low, Moderate, and High.

The feature set is organized into four broad categories:

1. Demographics and Heredity

- age: patient age, in years
- gender: patient sex (Female, Male, Other)
- family_hair_fall_history: binary indicator (0 = No, 1 = Yes) of hereditary predisposition to hair loss

2. Continuous Physiological and Serum Biomarkers

- Nutritional and mineral markers: total_protein (serum total protein, g/dL), iron (serum iron, µg/dL), calcium (serum calcium, mg/dL), vitamin_d (serum 25-hydroxy vitamin D, ng/mL), manganese (whole blood manganese, µg/L)
- Hair and hydration indices: total_keratine and hair_texture, both composite 0–100 scores reflecting hair condition rather than standard clinical laboratory measurements; body_water_content, expressed as a percentage
- Hepatic and stress biomarkers: alt_liver (serum alanine aminotransferase, U/L), stress_level (modeled on the 0–40 Perceived Stress Scale)

3. Clinical and Coexisting Conditions (binary flags: 0 = No, 1 = Yes)

- chronic_illness: presence of a pre-existing systemic medical condition
- anemia: documented iron or hemoglobin deficiency
- stress: self-reported experience of high psychological stress

4. Lifestyle and Environmental Exposures (binary flags: 0 = No, 1 = Yes)

- late_night_sleep: habitual late sleep timing or delayed circadian pattern
- sleep_disturbance: interrupted sleep or insomnia
- water_reason: regular exposure to hard or chemically treated water
- chemical_use: exposure to hair dyes, relaxers, or other chemical hair treatments

The dataset was constructed with domain-informed relationships linking these clinical and lifestyle variables to hair fall outcomes, reflecting established dermatological associations rather than being drawn from a single pre-existing clinical registry.

3.3 Data Preprocessing

To ensure data integrity and consistent handling across all three modeling architectures, the following preprocessing steps were applied prior to training:

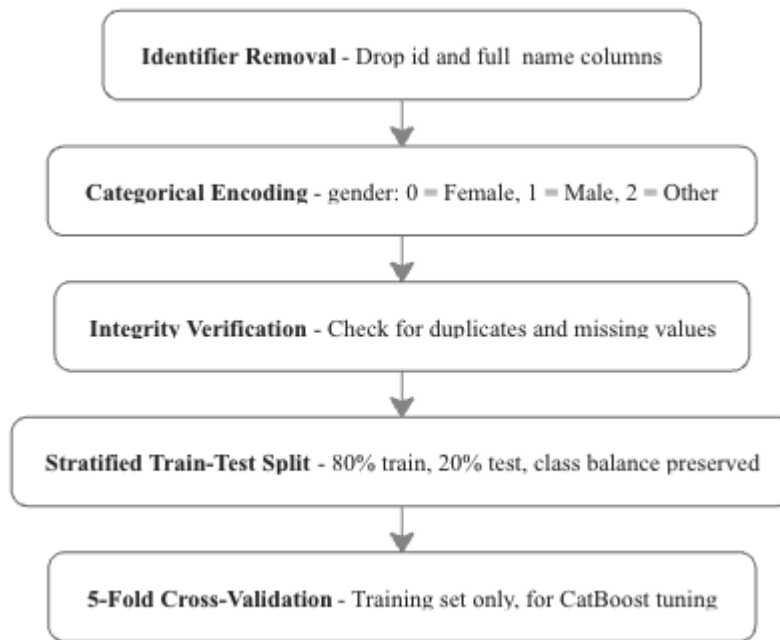


Figure 2: Data Preprocessing Pipeline

1. **Identifier Removal:** Non-informative identifier columns (id and full_name) were removed prior to training, as they carry no predictive value and risk introducing spurious associations if retained.
2. **Categorical Encoding:** The gender attribute was normalized and mapped into a numeric format (0 = Female, 1 = Male, 2 = Other) to accommodate models requiring numeric input. The target variable, hair_fall, was already represented as zero-indexed ordinal labels ($Y \in \{0, 1, 2\}$, corresponding to Low, Moderate, and High risk respectively), requiring no additional transformation.
3. **Missing Value Handling:** The dataset was verified to contain no missing values across any feature or target column following data cleaning, so no imputation step was required at the modeling stage.
4. **Train-Test Partitioning:** The dataset was partitioned using an 80/20 stratified train-test split, preserving the original class distribution across Low, Moderate, and

High risk tiers in both partitions. Stratified 5-Fold Cross-Validation was applied to the training partition only, used specifically for CatBoost's hyperparameter search; final performance metrics for all three models were computed on the held-out 20% test partition, which remained unseen during any training or tuning step.

5. Model-Specific Input Handling:

- For CatBoost: categorical and binary variables (gender, family_hair_fall_history, and related flags) were passed as numeric-encoded inputs alongside continuous biomarkers.
- For TabPFN and TabFM: continuous and binary features were passed directly to each model's attention mechanism without additional transformation, consistent with both models' design for handling raw, heterogeneous tabular inputs through in-context learning.

3.4 Model Selection and Rationale

Rather than relying on traditional baseline classifiers or simple ensemble voting schemes, this study benchmarks three distinct, state-of-the-art tabular modeling paradigms, each representing a different approach to structured data prediction.

3.4.1 State-of-the-Art Categorical Boosting: CatBoost

CatBoost was selected for its specialized handling of heterogeneous tabular data containing a mix of categorical and continuous variables. Unlike conventional tree-boosting algorithms, CatBoost mitigates target leakage through Ordered Target Encoding and constructs Symmetric (Oblivious) Decision Trees, which apply identical split conditions across all nodes at a given depth. The multi-class cross-entropy loss function minimized during training is defined as:

$$\mathcal{L}_{\text{MultiClass}} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=0}^2 y_{i,k} \log(\hat{p}_{i,k})$$

where $y_{i,k}$ is a binary indicator denoting whether class k is the correct label for instance i , and $\hat{p}_{i,k}$ is the model's predicted probability for that class. CatBoost's

regularized, symmetric tree structure provides a strong baseline against which the zero-shot foundation models can be meaningfully compared.

3.4.2 Frontier Tabular Foundation Model: TabPFN (v3)

TabPFN was selected as a representative, well-established tabular foundation model, offering zero-shot, prior-fitted prediction through in-context learning. Rather than undergoing dataset-specific training, TabPFN is pretrained entirely offline on synthetic causal prior distributions, and evaluates predictions directly through a single forward pass. Given a training context $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n$ and validation queries x_{test} , TabPFN computes the posterior predictive distribution as:

$$P(y_{\text{test}} \mid x_{\text{test}}, \mathcal{D}_{\text{train}}) = \text{TabPFN}_{\text{v2.5}}(x_{\text{test}}, \mathcal{D}_{\text{train}})$$

3.4.3 Zero-Shot Tabular Foundation Model: TabFM

TabFM was selected as a second, independently developed tabular foundation model, allowing this study to compare not only gradient boosting against foundation models broadly, but also two competing foundation model architectures against one another. Released by Google Research, TabFM combines alternating row- and column-wise attention with a dedicated in-context learning transformer, distinguishing its architectural approach from TabPFN's primarily sample-wise attention mechanism. Like TabPFN, TabFM requires no dataset-specific training or hyperparameter tuning, computing predictions as:

$$P(y_{\text{test}} \mid x_{\text{test}}, D_{\text{train}}) = \text{TabFM}(x_{\text{test}}, D_{\text{train}})$$

Including TabFM alongside TabPFN allows this study to assess whether performance gains attributed to zero-shot tabular foundation models generalize across independently developed architectures, or whether they are specific to a single model's design choices.

3.5 Experimental Setup and Hyperparameter Tuning

Model configurations were structured to reflect the distinct nature of each architecture: a tuned gradient-boosted model versus two zero-shot foundation models requiring no dataset-specific optimization.

- **CatBoost Setup:** Hyperparameter tuning was conducted via grid search across tree depth ($\text{depth} \in \{4, 6, 8\}$) and learning rate ($\text{learning_rate} \in \{0.01, 0.03, 0.1\}$), combined with L2-regularized leaf weights. Early stopping was applied after 50 rounds without improvement to prevent overfitting. The optimal configuration was selected using Stratified 5-Fold Cross-Validation on the training partition, based on macro-F1 score.
- **TabPFN Setup:** As a pretrained foundation model operating through in-context learning, TabPFN required no hyperparameter tuning or gradient-based optimization. Training data was supplied directly as context, and predictions for the held-out test set were generated in a single forward pass.
- **TabFM Setup:** Consistent with its zero-shot design, TabFM likewise required no dataset-specific hyperparameter search. The training partition was supplied as in-context data, and test predictions were generated directly through TabFM's row-and-column attention mechanism without any gradient descent or fine-tuning step.

This asymmetry in tuning effort extensive grid search for CatBoost versus no tuning at all for TabPFN and TabFM is intentional and central to this study's research question: whether zero-shot foundation models can match or exceed a carefully tuned gradient-boosted baseline without requiring comparable optimization effort.

3.6 Explainability Architecture (SHAP Integration)

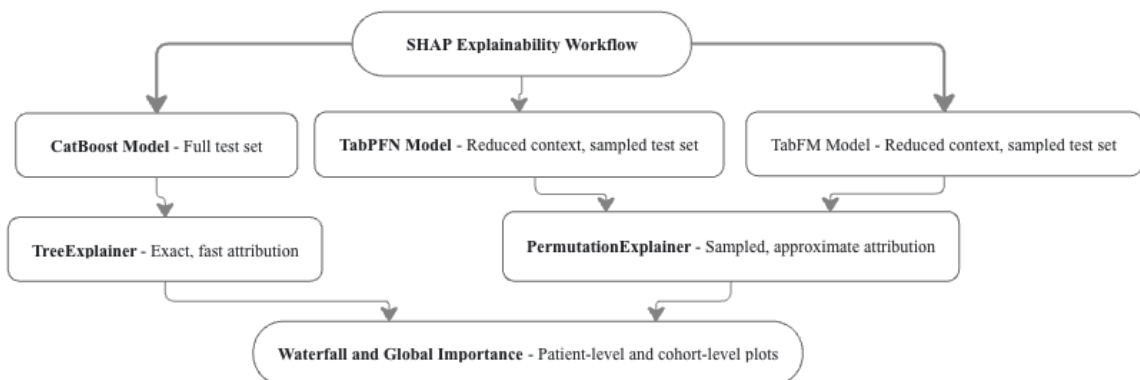


Figure 3: SHAP Explainability Workflow

To address the interpretability limitations common to both boosting and transformer-based architectures, post-hoc explainability was integrated across all three trained models using Shapley Additive Explanations (SHAP). Grounded in cooperative game theory, SHAP decomposes each model's prediction into additive per-feature contributions:

$$f(x) = \phi_0 + \sum_{j=1}^M \phi_j$$

where ϕ_0 is the baseline expected prediction and ϕ_j quantifies the specific contribution of feature j (for example, iron, stress_level, total_protein, or family_hair_fall_history) to the final predicted risk tier.

For CatBoost, SHAP values were computed using the TreeExplainer algorithm, which exploits the model's tree structure to compute exact Shapley values efficiently across the full test set. For TabPFN and TabFM, SHAP values were instead computed using PermutationExplainer, since neither model exposes an internal tree structure that TreeExplainer can exploit. Because permutation-based SHAP estimation requires repeated forward passes through each model's full in-context attention mechanism, its computational cost scales substantially with both context size and the number of samples explained. To keep explainability computation tractable, TabPFN's and TabFM's SHAP analyses were each conducted using a reduced training context and a sampled subset of test instances, rather than the full training set and complete test partition used for TreeExplainer. This methodological difference reflects a genuine computational constraint of permutation-based explainability on transformer architectures, not a difference in the underlying explanatory approach.

Across all three models, this framework produces both patient-level waterfall plots, showing how individual features shifted a specific prediction away from the baseline, and cohort-level global importance rankings, showing which features most consistently influenced predictions across the dataset as a whole.

3.7 Tools and Technologies

This research was implemented using the following Python-based software stack:

- **Data Manipulation and Preprocessing:** Pandas and NumPy for data handling, and Scikit-learn for stratified train-test splitting, Stratified 5-Fold Cross-Validation, and standard evaluation metrics.
- **Tabular AI Engines:** The official CatBoost framework for gradient-boosted tree training; the TabPFN Python package for zero-shot tabular foundation model inference; and Google Research's TabFM package for the second zero-shot tabular foundation model.
- **Explainable AI:** The official SHAP library, used to compute both TreeExplainer-based and PermutationExplainer-based feature attributions across all three models.
- **Statistical Analysis:** Statsmodels, used to compute McNemar's test for pairwise statistical significance testing between model predictions, and Wilson score confidence intervals for reported accuracy figures.

3.8 Experimental Environment

All experiments were executed within a standardized, cloud-based computing environment to ensure consistent execution time and latency measurements across all three models.

Table 1: Experimental Hardware and Software Specifications

Category	Parameter	Hardware / Software Specification
Hardware	System Environment	Google Colaboratory (cloud-based)
Hardware	Secondary Environment	Kaggle Notebooks (cloud-based)
Hardware	System Memory (RAM)	~13 GB (Colab) / ~30 GB (Kaggle)
Hardware	GPU Accelerator	NVIDIA T4 GPU (Colab: single T4; Kaggle: dual T4, single GPU used)
Hardware	Operating System	Ubuntu 22.04.5 LTS (Jammy Jellyfish)
Software	Programming Language	Python 3.x
Software	Acceleration Engine	PyTorch (CUDA, NVIDIA T4 GPU)

Software	Core AI & ML Libraries	CatBoost, TabPFN, TabFM, SHAP, Scikit-learn, Pandas, NumPy, Statsmodels
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The use of a cloud-based GPU environment was necessary to support the computational demands of TabPFN and TabFM, both of which rely on GPU-accelerated attention mechanisms for tractable inference times, particularly given the dataset's scale of 21,606 records.

3.9 Performance Evaluation Metrics

Model performance across the three ordinal risk categories (Low, Moderate, High) was evaluated using macro-averaged statistical metrics, applied consistently across all three models, alongside computational efficiency indicators.

- **Accuracy:** Quantifies overall classification accuracy across all three risk tiers:

$$\text{Accuracy} = \frac{\sum_{k=0}^2 TP_k}{N}$$

- **Macro-Precision:** Evaluates prediction fidelity across classes equally, without weighting by class size:

$$\text{Precision}_{\text{Macro}} = \frac{1}{3} \sum_{k=0}^2 \frac{TP_k}{TP_k + FP_k}$$

- **Macro-Recall:** Measures sensitivity across target levels equally:

$$\text{Recall}_{\text{Macro}} = \frac{1}{3} \sum_{k=0}^2 \frac{TP_k}{TP_k + FN_k}$$

- **Macro-F1 Score:** Computes the harmonic mean of macro-precision and macro-recall:

$$\text{Macro-F1} = 2 \times \frac{\text{Precision}_{\text{Macro}} \times \text{Recall}_{\text{Macro}}}{\text{Precision}_{\text{Macro}} + \text{Recall}_{\text{Macro}}}$$

- **ROC-AUC:** Computed using a one-vs-rest strategy across the three risk classes, macro-averaged to summarize discriminative performance independent of class size.

- **Computational Latency:** Training or context-fitting wall-clock time (seconds) and per-sample inference latency (milliseconds/sample) were recorded for each model to evaluate deployment feasibility, given that CatBoost's training-then-predict paradigm differs fundamentally from TabPFN's and TabFM's in-context, fit-free inference approach.
- **Statistical Significance Testing:** Pairwise differences in prediction accuracy between models were assessed using McNemar's test on the held-out test set, with 95% confidence intervals for each model's accuracy computed via the Wilson score interval method, providing a statistically grounded basis for comparing model performance beyond raw metric values alone.

Chapter 4: Expected Outcome and Working Schedule

4.1 Expected Outcomes

This study aims to systematically evaluate and compare the performance of three distinct tabular AI paradigms: CatBoost, a gradient-boosted decision tree architecture with ordered categorical encoding; TabPFN, a zero-shot prior-fitted transformer foundation model; and TabFM, Google Research's more recently released zero-shot tabular foundation model built on a hybrid row-and-column attention architecture for accurate, multi-tier hair fall risk classification across 21,606 patient records.

By benchmarking CatBoost against both TabPFN and TabFM, this research assesses not only whether zero-shot, in-context transformer learning can match or outperform a carefully tuned gradient-boosted baseline, but also whether such performance characteristics are consistent

across two independently developed foundation model architectures, or specific to a single model's design.

The embedded Explainable AI (XAI) layer, implemented through Shapley Additive Explanations (SHAP) and applied uniformly across all three models, provides comprehensive feature-level transparency, transforming complex model outputs into an interpretable clinical decision-support asset. By generating both cohort-level feature rankings and patient-specific waterfall plots for each of the three models, the system clarifies how compounding physiological indicators (such as iron, total_protein, vitamin_d, and calcium) and behavioral or clinical factors (stress_level, family_hair_fall_history, anemia, chronic_illness) directly drive individual risk assignments across the Low, Moderate, and High risk tiers.

Consequently, the final framework is expected to minimize misclassification errors while delivering reliable, objective risk predictions, and to clarify the comparative strengths and trade-offs in predictive performance, computational latency, and explainability of gradient boosting versus two competing zero-shot tabular foundation model architectures. Combining strong multi-class predictive metrics (Macro-F1, Accuracy, ROC-AUC) with statistically grounded model comparison (via McNemar's test) and explicit feature attribution tracking is intended to help individuals recognize early risk indicators, while providing a transparent, data-driven decision-support reference for further clinical interpretation.

4.2 Working Schedule

The research was carried out across an extended timeline beyond the originally proposed 12-week window, reflecting additional iterations required during dataset validation and preprocessing, as well as the subsequent extension of the study to incorporate a third benchmarked model, TabFM, following its public release in mid-2026. Table 2 presents a week-by-week breakdown of core project milestones, distinguishing completed phases from work carried out in later stages of the project.

Table 2: Week-by-Week Project Gantt Chart (May 2026 – July 2026)

Core Activities	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
1. Problem Formulation & Domain Study	■	■										

2. Research Objectives & Framework Design		■	■									
3. Literature Review & Reference Mapping			■	■								
4. Dataset Cleaning & Stratified Preprocessing				■	▨	▨						
5. Model Setup (CatBoost & TabPFN v3)						▨	▨	▨				
6. Hyperparameter Tuning & In-Context Setup								▨	▨			
7. Cross-Validation & Model Benchmarking									▨	▨		
8. Explainability (SHAP Integration & Plots)										▨	▨	
9. Metrics Evaluation & Performance Analysis											▨	▨
10. Final Thesis Documentation & Presentation												▨

■	Completed (Work Done)
▨	In Progress (Work Remaining)

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